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  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "09-07-2026_17:43:53",
  "variant" : 3,
  "subjectName" : "mathematics",
  "categories" : [ "Probability", "Combinatorics", "Mathematical Logic" ],
  "questions" : [ {
    "question" : "What is the probability of flipping a fair coin three times and getting all heads?",
    "answers" : [ "1/6", "1/2", "1/4", "1/8" ],
    "correctIndex" : 3
  }, {
    "question" : "What is the sum of probabilities of all possible mutually exclusive outcomes in a sample space?",
    "answers" : [ "Infinite", "0", "100", "1" ],
    "correctIndex" : 3
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    "question" : "In how many ways can 4 people be seated in a straight line?",
    "answers" : [ "24", "6", "16", "12" ],
    "correctIndex" : 0
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    "question" : "What is the value of the combinatorial coefficient  $C(5, 2)$ ?",
    "answers" : [ "10", "5", "15", "20" ],
    "correctIndex" : 0
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    "question" : "How many unique permutations can be made from the letters in the word 'APP'?",
    "answers" : [ "3", "6", "9", "1" ],
    "correctIndex" : 0
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    "question" : "How many distinct subsets can be formed from a set containing 5 elements?",
    "answers" : [ "32", "64", "25", "16" ],
    "correctIndex" : 0
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    "question" : "If P is true and Q is false, what is the truth value of the implication  $P \rightarrow Q$ ?",
    "answers" : [ "False", "Contingent", "True", "Undefined" ],
    "correctIndex" : 0
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    "question" : "What is a logical statement called if it evaluates to true under all possible truth value assignments?",
    "answers" : [ "Contingency", "Fallacy", "Tautology", "Contradiction" ],
    "correctIndex" : 2
  }, {
    "question" : "What is the negation of the statement 'All primes are odd'?",

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"answers" : [ "Some primes are odd", "There exists at least one prime that is not odd", "No primes  
are odd", "All primes are even" ],  
"correctIndex" : 1  
}, {  
"question" : "Which logical connective represents an 'exclusive or' (XOR)?",  
"answers" : [ "Both inputs are false", "Exactly one input is true", "Both inputs are true", "At least  
one input is true" ],  
"correctIndex" : 1  
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